Predicting students' performance

I. Introduction:

Predicting student performance is a key issue in educational data analysis for policymakers to identify learning difficulties and improve academic outcomes. In fact, to enhance their teaching abilities, teachers must be able to forecast pupils' performance tendencies. In our project, we use RStudio to analyze (cleaning, classification, and regression) the *Student Performance* dataset, which provides an extensive array of data on students' demographics, family background, school-related variables, and past academic success. Our main goal is to understand how these factors explain students' final performance in mathematics and to develop models that predict the probability of passing the final exam.

We begin by cleaning the data by checking for missing values, duplicates, and extreme values. We classify our data by categorizing relevant variables into a binary indicator (pass or fail) and a five-level performance category (A-F scale). Then, we create dummy variables for multi-category factors such as parental job type and reasons for choosing a school, allowing their integration into regression models.

After cleaning and classifying the data, we perform descriptive statistics, correlation matrices, and several linear regression analyses to evaluate how each explanatory variable individually influences final performance.

Finally, we compute multiple linear regression models under two contexts. The first model focuses on academic performance and considers past grades, failures, absences, and study time. The second one is a socio-behavioral model that incorporates parental education, weekend and weekday alcohol usage, and other behavioral factors.

These models allow us to compare academic determinants with social and behavioral determinants of school success. We additionally construct a binary variable indicating whether the student passed or failed that enables us to do a logistic regression. This project seeks to better understand the critical factors influencing academic achievement and to forecast students' final grades.

II. Programming tools and Data analysis framework in R :

In our analysis, we use RStudio because of its flexibility in econometric modelling, data manipulation, and visualization. We analyze a dataset called *student_mat_csv*, which we extracted from the "Student Performance" dataset, providing thorough information on students enrolled in a mathematics course. The dataset includes variables grouped into the following categories: Demographic variables (`age`, `sex`, `address`, `famsize`, `Pstatus`), family background (`Medu`, `Fedu`, `Mjob`, `Fjob`, `guardian`), school-related resources (`schoolsup`, `famsup`, `paid`, `activities`, `higher`, `internet`, `nursery`), behavioral variables (`goout` “going out time”, Dalc, Walc, health, freetime, traveltime), academic performance variables (G1, G2, G3 (first, second, and final period grades), studytime, failures, absences). Our goal is to develop predictive models of academic achievement and to understand the factors that influence students' final grades (G3).

The analysis starts with importing data and inspecting it, using functions such as read_excel, str, and summary to load the dataset and understand its structure. This initial step allows us to identify variable types and potential data quality issues. After loading the dataset, we performed several cleaning steps:

- inspection of variable structure using str();
- identification of missing values with colSums(is.na(.));
- removal of incomplete rows using complete.cases();
- detection and removal of duplicate observations;
- conversion of categorical variables (e.g., *sex*, *school*, *address*, *Pstatus*, *schoolsup*) into R factors.

To simplify the dataset, we created a binary pass/fail variable (G3_binary), where a student passes if G3 ≥ 10 using ifelse, and a categorical performance scale (G3_5cat) using the cut function. With the help of RStudio, we were able to transform the data so we can intuititvely analyze it. We looked at outliers, especially in the absence variable. Students with more than 50 absences were filtered out in order to cap extremely high absenteeism scores. Complete, consistent, and organized observations are included in the final cleaned dataset, making it appropriate for both statistical modeling and exploration.

To explore the data, we use descriptive statistics and graphical tools such as table, hist, boxplot, and plot. Then, for regression, we used the lm function for linear regression and the glm function for logistic regression, demonstrating how RStudio intuitively combines programming and statistical analysis.

The preprocessed student related variables :

Attribute	**Domain (description)**
sex	student’s sex (binary: female or male)
age	student’s age (numeric: from 15 to 22)
school	student’s school (binary: *Gabriel Pereira* or *Mousinho da Silveira*)
address	student’s home address type (binary: urban or rural)

Pstatus	parent's cohabitation status (binary: living together or apart) Medu
Mjob	mother's job (nominal[b])
Fedu	father's education (numeric: from 0 to 4[a])
Fjob	father's job (nominal[b])
guardian	student's guardian (nominal: mother, father or other)
famsize	family size (binary: ≤ 3 or > 3)
famrel	quality of family relationships (numeric: from 1 – very bad to 5 – excellent)
reason	reason to choose this school (nominal: close to home, school reputation, course preference or other)
traveltime	home to school travel time (numeric: 1 – < 15 min., 2 – 15 to 30 min., 3 – 30 min. to 1 hour or 4 – > 1 hour).
studytime	weekly study time (numeric: 1 – < 2 hours, 2 – 2 to 5 hours, 3 – 5 to 10 hours or 4 – > 10 hours)
failures	number of past class failures (numeric: n if $1 \leq n < 3$, else 4)
schoolsup	extra educational school support (binary: yes or no)
internet	Internet access at home (binary: yes or no)
studytime	weekly study time (numeric: 1 – < 2 hours, 2 – 2 to 5 hours, 3 – 5 to 10 hours or 4 – > 10 hours)
famsup	family educational support (binary: yes or no)

schoolsup	extra educational school support (binary: yes or no)
internet	Internet access at home (binary: yes or no)
nursery	attended nursery school (binary: yes or no)
freetime	free time after school (numeric: from 1 – very low to 5 – very high)
goout	going out with friends (numeric: from 1 – very low to 5 – very high)
nursery	attended nursery school (binary: yes or no) higher
freetime	free time after school (numeric: from 1 – very low to 5 – very high)
goout	going out with friends (numeric: from 1 – very low to 5 – very high)
Walc	weekend alcohol consumption (numeric: from 1 – very low to 5 – very high)
Dalc	workday alcohol consumption (numeric: from 1 – very low to 5 – very high)
health	current health status (numeric: from 1 – very bad to 5 – very good) absences
absences	number of school absences (numeric: from 0 to 93)
G1	first period grade (numeric: from 0 to 20)
G2	second period grade (numeric: from 0 to 20)
G3	final grade (numeric: from 0 to 20)

III. Exploratory Data Analysis (EDA):

1. **Descriptive statistics:**

The study uses data from a math performance dataset comprising 395 cases and 33 features related to students' academic results, as well as personal, household, economic, and conduct factors. We conduct a detailed descriptive analysis before econometric estimation to characterize the population studied, verify the consistency of the variables, and guide the choice of econometric specifications.

- **<u>Academic variables :</u>**

School grades (G1, G2, G3) : The variables G1, G2 and G3 represent the grades for the first, second and third terms, respectively, marked out of 20.

```
> summary(student_mat$G1)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   3.00    8.00   11.00   10.92   13.00   19.00
> summary(student_mat$G2)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00    9.00   11.00   10.72   13.00   19.00
> summary(student_mat$G3)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00    8.00   11.00   10.42   14.00   20.00
>
```

With a small decline between G1 and G3, the means of the three grades are close to the pass mark (10/20). This pattern suggests either increased selectivity in the final exam or rising difficulty throughout the academic year. On the other hand, we observe a wide range of grades (from 0 to 20 for G3), demonstrating a considerable dispersion in student performance. This indicates significant heterogeneity among students, implying that they respond differently to academic demands, family background, and behavioral factors.

- **<u>Academic effort and time constraints :</u>**

The variables weekly studytime (numeric: 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - >10 hours) and travel time from home to school (numeric: 1 - <15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - >1 hour)are measured as ordinal categories.

```
> summary(student_mat$studytime)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   1.000   2.000   2.036   2.000   4.000
> summary(student_mat$traveltime)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   1.000   1.000   1.449   2.000   4.000
>
```

We observe that most students show moderate study effort and short travel times. The mean of study time is nearly 2, demonstrating that only a small number of students study intensively. Travel time has a median of 1, indicating that the majority of students live within 15 minutes of school. These trends show that time constraints related to travel are limited for most students, whereas study effort varies modestly and may help explain differences in academic performance.

- **<u>Absences and treatment of extreme values:</u>**

```
> # Extreme values:
>
> student_mat <- student_mat %>% filter(absences > 50)
> summary(student_mat$absences)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  54.00   55.00   56.00   61.67   65.50   75.00
>
```

We observe extreme values (up to 75 absences), meaning that a small percentage of students exhibit abnormal behavior. Econometric estimations may be unduly influenced by these values, which is why we eliminate observations with absences of more than fifty.

- **<u>Family background :</u>**

The Medu and Fedu variables measure the level of education of the mother and father (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education).

```
> summary(student_mat$Medu)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      3       3       3       3       3       3
> summary(student_mat$Fedu)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.000   2.500   3.000   2.667   3.000   3.000
>
```

Average levels of parental education are intermediate, and there is a strong correlation between the mother's and father's education.

- **<u>Behavioural variables :</u>**

The variables observed are weekday alcohol consumption (Dalc), weekend alcohol consumption (Walc) and health (numeric: from 1 - very bad to 5 - very good).

```
> summary(student_mat$Dalc)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   1.500   2.000   1.667   2.000   2.000
> summary(student_mat$Walc)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   2.000   3.000   2.667   3.500   4.000
> summary(student_mat$health)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   3.000   5.000   3.667   5.000   5.000
>
```

Students show generally good health and minimal alcohol intake. With a mean of 1.67 and a median of 2, weekday alcohol consumption (Dalc) is modest, suggesting less drinking during the week. On the other hand, weekend alcohol consumption (Walc) is greater, with a mean of 2.67 and a median of 3, indicating more drinking on weekends. With a mean score of 3.67 and a median score of 5, health surveys indicate that students have a favorable opinion of their health.
The variability of these variables justifies their inclusion in socio-behavioural and logit models.

- **Academic success (Pass/Fail) :**

```
> student_mat$PassFail <- ifelse(student_mat$G3 >= 10, "Pass", "Fail")
> student_mat$PassFail <- factor(student_mat$PassFail, levels = c("Pass", "Fail"))
> table(student_mat$PassFail)

Pass Fail
 265  130
> prop.table(table(student_mat$PassFail))

     Pass      Fail
0.6708861 0.3291139
>
```

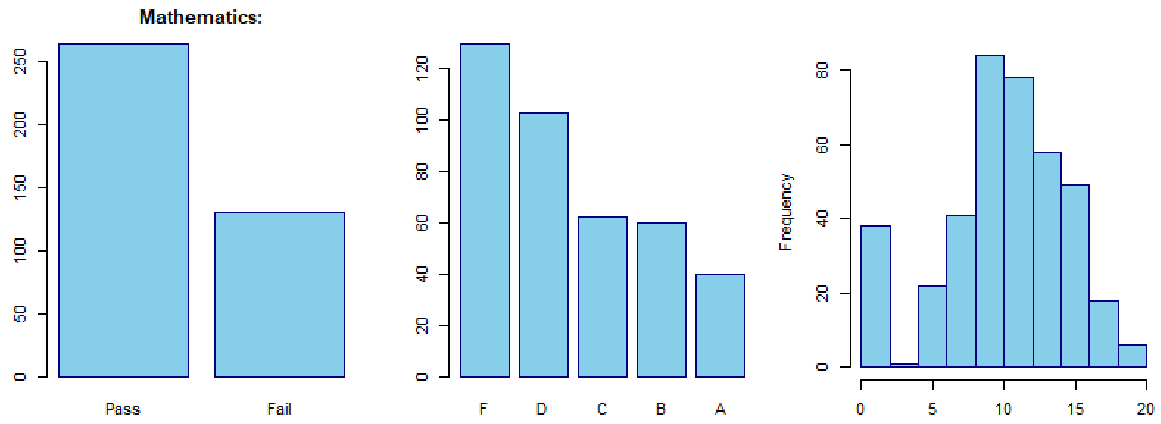
We construct from the final grade (G3) a binary pass/fail variable indicating that roughly two-thirds (67,1%) of students pass the academic year. This analysis allows us to fit a logit model to describe the probability of success.

```
>
> table(student_mat$G3_cat5)

  F   D   C   B   A
130 103  62  60  40
>
```

The distribution of the five-category G3 shows that although many students pass, only a few attain top grades, indicating heterogeneity in academic achievement. This asymmetry is consistent with a system that involves selective assessment.

2. Graphical initial analysis of students' academic performance :

We use graphical representations of students' academic performance in mathematics to complement the descriptive statistics. The following graphs help us better understand the distribution of results and identify important trends that drive the subsequent econometric analysis.



The graphical analysis emphasizes what we observed previously with three key insights:

1. There is a distinct difference between pupils who pass and those who fail in terms of academic results.
2. Classification models are especially important because performance is measured against the passing threshold.
3. The use of linear regression models to examine performance factors is justified by the significant dispersion in final grades.

3. **Correlation analysis :**

- **<u>Purpose :</u>**

Before estimation, we compute a crucial methodological step, which is computing correlations between the final grade in mathematics (G3) and the explanatory variables. This step helps us determine if the explanatory variables are positively, negatively, or weakly related to academic performance. Strong correlations with G3 indicate suitability for simple linear regressions (SLR), suggesting that they explain significant variations in grades. Weaker correlations may still hold theoretical relevance but are likely to have limited explanatory power unless controlled for in multiple linear regressions (MLR).

❖ **<u>Key correlations :</u>**

Some of the key variables and their correlations with G3 :

Variable	**Corr (G3)**
G2	**0.90**
G1	**0.80**
failures	**-0.36**
studytime	0.10

absences	0.03
Medu	0.22
Fedu	0.15

- **Implications for modelling :**

Regarding academic performance variables, **G2** shows a very strong positive correlation with G3 ; students who perform well in the second period tend to perform well in the final evaluation. **G1** is also highly and positively correlated with G3, but less than G2. This shows the effect of early academic achievement on later success. **Failures** capture cumulative academic difficulties not fully reflected in current grades. It is negatively correlated with G3.

For effort and attendance variables, **studytime** is theoretically relevant and may have a conditional effect once other factors are controlled for. It displays a weak but positive correlation with G3. Also, **absences** can be used as a standard proxy for engagement and discipline, and their effect may emerge in multivariate or nonlinear models. They show a very small correlation with G3.

As for family background, **Mother's education (Medu)** has a positive correlation with G3, which means that higher parental education, particularly that of the mother, is associated with better academic outcomes. On the other hand, **Father's education (Fedu)** also indicates a positive but weaker correlation with G3.

4. Simple Linear Regressions:

We begin with multiple simple linear regressions taking the final grade G3 as the dependent variable and regressing it on one explanatory variable at a time. This step allows us to determine the marginal effect and know the intuition behind each potential explanatory factor. SLR models take the form: $\mathbf{G3 = \alpha + \beta X_i + u_i}$ **.**

- **Final Grade and First-Period Grade (G3 ~ G1)**

In the first model, we regress the final grade on the first-period grade. From the table, we see that the effect of G1 on G3 is positive, as expected, and statistically significant. A one-point increase in G1 increases G3 by 1.1 points. This solid link shows that students who perform well early tend to maintain their advantage. Still, the estimated coefficient above one suggests that early scores might reflect unobserved factors, such as ability, leading to unobserved heterogeneity.

```
Call:
lm(formula = G3 ~ G1, data = student_mat)

Residuals:
     Min       1Q   Median       3Q      Max
-11.6223  -0.8348   0.3777   1.6965   5.0153

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.65280    0.47475  -3.481 0.000555 ***
G1           1.10626    0.04164  26.568  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.743 on 393 degrees of freedom
Multiple R-squared:  0.6424,	Adjusted R-squared:  0.6414
F-statistic: 705.8 on 1 and 393 DF,  p-value: < 2.2e-16
```

- **Final Grade and Second-Period Grade (G3 ~ G2)**

In the second model, we regress G3 on G2. The latter is closer to the final grade, so it likely predicts results better, as shown by the very high R^2 (82%). As expected, G2 positively affects the dependent variable and has nearly the same effect as G1. This finding shows that current grades matter most when predicting final outcomes. Much better than earlier data from G1, because newer information packs more clues about what's coming.

```
Call:
lm(formula = G3 ~ G2, data = student_mat)

Residuals:
    Min      1Q  Median      3Q     Max
-9.6284 -0.3326  0.2695  1.0653  3.5759

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.39276    0.29694   -4.69 3.77e-06 ***
G2           1.10211    0.02615   42.14  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.953 on 393 degrees of freedom
Multiple R-squared:  0.8188,	Adjusted R-squared:  0.8183
F-statistic:  1776 on 1 and 393 DF,  p-value: < 2.2e-16
```

- **Final Grade and Past Failures (G3 ~ failures)**

In this model, we check how G3 is linked to past class failures. This factor indicates ongoing problems, suggesting persistent knowledge gaps or fading interest in classes. The effect is statistically significant and, as expected, negative. A one-point increase in failure decreases the final grade by 2.2 points. Intuitively, students who have failed in the past face structural disadvantages that persistently affect their performance, even if their current grades improve.

```
Call:
lm(formula = G3 ~ failures, data = student_mat)

Residuals:
     Min       1Q   Median       3Q      Max 
-11.1572  -2.1572  -0.1572   2.8428   9.0632 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  11.1572     0.2361   47.26  < 2e-16 ***
failures     -2.2204     0.2899   -7.66 1.47e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.279 on 393 degrees of freedom
Multiple R-squared:  0.1299,	Adjusted R-squared:  0.1277 
F-statistic: 58.67 on 1 and 393 DF,  p-value: 1.466e-13
```

- **Final Grade and Mother's Education (G3 ~ Medu)**

Since parental education is commonly used as a proxy for educational support at home, we compute this model by regressing G3 on the mother's level of education. The effect of Medu on the dependent variable is statistically significant, indicating that a one-point increase in Medu increases G3 by approximately 0.9 points. However, R^2 is very low (5%), indicating that the explanatory power of family background is limited when considered alone. This could mean that the variable affects students' performance indirectly rather than directly.

```
Call:
lm(formula = G3 ~ Medu, data = student_mat)

Residuals:
     Min       1Q   Median       3Q      Max 
-11.5517  -1.7342   0.4483   3.2202   9.2658 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   7.9167     0.6097   12.98  < 2e-16 ***
Medu          0.9088     0.2061    4.41 1.34e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.478 on 393 degrees of freedom
Multiple R-squared:  0.04715,	Adjusted R-squared:  0.04473 
F-statistic: 19.45 on 1 and 393 DF,  p-value: 1.336e-05
```

- **Final Grade and Study Time (G3 ~ studytime)**

In the last SLR model, we regress the final grade on study time per week. This variable is a central factor in educational production functions. The estimated coefficient is positive but only marginally significant. The limited explanatory power indicates that the quantity of study alone is insufficient, even though longer study sessions are associated with somewhat higher scores. The impact of raw study hours is probably dominated by variations in prior ability, study quality, and efficiency.

```
Call:
lm(formula = G3 ~ studytime, data = student_mat)

Residuals:
     Min       1Q   Median       3Q      Max 
-11.4643  -1.8623   0.5357   3.0697   9.1377 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   9.3283     0.6033  15.463   <2e-16 ***
studytime     0.5340     0.2741   1.949   0.0521 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.565 on 393 degrees of freedom
Multiple R-squared:  0.009569,	Adjusted R-squared:  0.007049 
F-statistic: 3.797 on 1 and 393 DF,  p-value: 0.05206
```

5. Multiple Linear Regressions :

- **The Academic Performance model :**

The model **G3 ~ G2 + G1 + failures + absences + studytime** helps us predict the final grade using the most informative school variables. It shows a very high R^2 (83%) which demonstrates high explanatory power of the variance in G3.

G2 has a positive and statistically significant effect on the dependent variable. If the second-period grade increases by 1 point, the final grade increases by nearly 1 point as well.

G1 remains positive, but its size drops significantly when G2 is added. This shows that current performance is more important than older results.

The variable **failures** have a negative and statistically significant effect on G3. Even after controlling for G1 and G2, students with past class failures have worse final grades.

The variables' **absences** and **study time** don’t have a stable effect on the final grade, indicating that their influence is mostly mediated by grades rather than exerting a direct impact.

```
Call:
lm(formula = G3 ~ G1 + G2 + failures + absences + studytime,
    data = student_mat)

Residuals:
    Min      1Q  Median      3Q     Max
-9.1894 -0.3662  0.2649  0.9706  3.6031

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.44276    0.43247  -3.336 0.000931 ***
G1           0.14996    0.05580   2.687 0.007510 **
G2           0.97726    0.04914  19.888  < 2e-16 ***
failures    -0.28377    0.14041  -2.021 0.043968 *
absences     0.03664    0.01205   3.040 0.002530 **
studytime   -0.17817    0.11717  -1.521 0.129177
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.908 on 389 degrees of freedom
Multiple R-squared:  0.8287,    Adjusted R-squared:  0.8265
F-statistic: 376.4 on 5 and 389 DF,  p-value: < 2.2e-16
```

- **The socio-behavioral model :**

In the following model, **G3 ~ failures + absences + studytime + Walc + Dalc + Medu + Fedu**, we go beyond grades and focus on factors such as family, environment, and effort related to the student.

Failures continue to be very detrimental and important, demonstrating the enduring impact of past academic defeats. When Medu is considered, the mother's education is positive and significant, whereas the father's education (Fedu) is often weaker or insignificant. Once family background and failure history are taken into account, study time and absences often have little explanatory power, and the consequences of alcohol intake are negligible and statistically insignificant. Because grades account for more variance than these factors, the R^2 is moderate.

```
Call:
lm(formula = G3 ~ failures + absences + studytime + Walc + Dalc +
    Medu + Fedu, data = student_mat)

Residuals:
     Min       1Q   Median       3Q      Max
-12.0168  -2.0499   0.3525   3.0284   8.1268

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.12142    1.01771   8.963  < 2e-16 ***
failures    -2.01597    0.30698  -6.567 1.65e-10 ***
absences     0.02456    0.02732   0.899   0.3694
studytime    0.18631    0.26823   0.695   0.4877
Walc         0.06903    0.22333   0.309   0.7574
Dalc        -0.12196    0.31798  -0.384   0.7015
Medu         0.62212    0.25554   2.435   0.0154 *
Fedu        -0.09514    0.25582  -0.372   0.7102
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.257 on 387 degrees of freedom
Multiple R-squared:  0.1519,	Adjusted R-squared:  0.1365
F-statistic:   9.9 on 7 and 387 DF,  p-value: 2.226e-11
```

- **The socio-economic model :**

In the socio-economic model, we focus on family and resource variables related to students' performance. These variables, when considered alone, have limited direct explanatory power. Mother's education is statistically significant in this MLR model, consistent with the literature on education and parental human capital. However, the model explains only 6% of the variance in G3, indicating that it doesn't capture the effect of the socio-economic variables well.

```
Call:
lm(formula = G3 ~ Medu + Fedu + famsize + internet + activities,
    data = student_mat)

Residuals:
     Min       1Q   Median       3Q      Max
-12.2011  -2.2011   0.5187   2.8312   9.3543

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.0818     0.7908   8.955  < 2e-16 ***
Medu          0.7956     0.2668   2.982  0.00305 **
Fedu          0.1405     0.2655   0.529  0.59707
famsizeLE3    0.9242     0.4975   1.858  0.06399 .
internetyes   0.6917     0.6161   1.123  0.26222
activitiesyes -0.1005    0.4537  -0.221  0.82486
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.472 on 389 degrees of freedom
Multiple R-squared:  0.0592,	Adjusted R-squared:  0.0471
F-statistic: 4.895 on 5 and 389 DF,  p-value: 0.0002342
```

- **Interaction model : academic persistence (G1) × past difficulty (failures) :**

In this model, we use an interaction term to examine whether performance in the first period is the same across students with a history of academic failure. This enables us to determine if the permanence of academic accomplishment is moderated by past failures.

We consider a null hypothesis, namely that past failures don't affect the impact of early performance (G1) on final outcomes (G3). The interaction term has a statistically insignificant coefficient, so we fail to reject the null hypothesis. The marginal effect of G1 on G3 is the same across students with and without past failures.

Given its significance, early academic achievement (G1) is the primary driver of outcomes among the other factors. Once early grades are taken into account, family background and effort factors lose their direct explanatory power.

```
Call:
lm(formula = G3 ~ G1 + failures + G1:failures + Medu + studytime,
    data = student_mat)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.6825  -0.8219   0.2678   1.6223   5.2354 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.17587    0.69370  -1.695   0.0909 .  
G1           1.07990    0.04761  22.681   <2e-16 ***
failures     0.08559    0.60347   0.142   0.8873    
Medu         0.16333    0.13020   1.254   0.2104    
studytime   -0.22461    0.16679  -1.347   0.1789    
G1:failures -0.07570    0.07067  -1.071   0.2848    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.715 on 389 degrees of freedom
Multiple R-squared:  0.6534,	Adjusted R-squared:  0.6489 
F-statistic: 146.7 on 5 and 389 DF,  p-value: < 2.2e-16
```

- **Logistic regression : probability of passing :**

To account for the binary variable we created from the dependent variable (1 if the student passes “G3≥10” and 0 if he fails “G3<10”), we estimate a logit model. The latter is very useful because it shows the probability of passing and failing.

From the table, we observe that the second-period grade has a strong, statistically significant effect on the probability of passing. These results confirm what we found earlier: G2 is very important for predicting the final grade, demonstrating the persistence of academic achievement.

The significance of the study time coefficient indicates that greater effort increases the probability of success on the final exam. Given the positive, significant coefficient of age, we conclude that more mature students perform better.

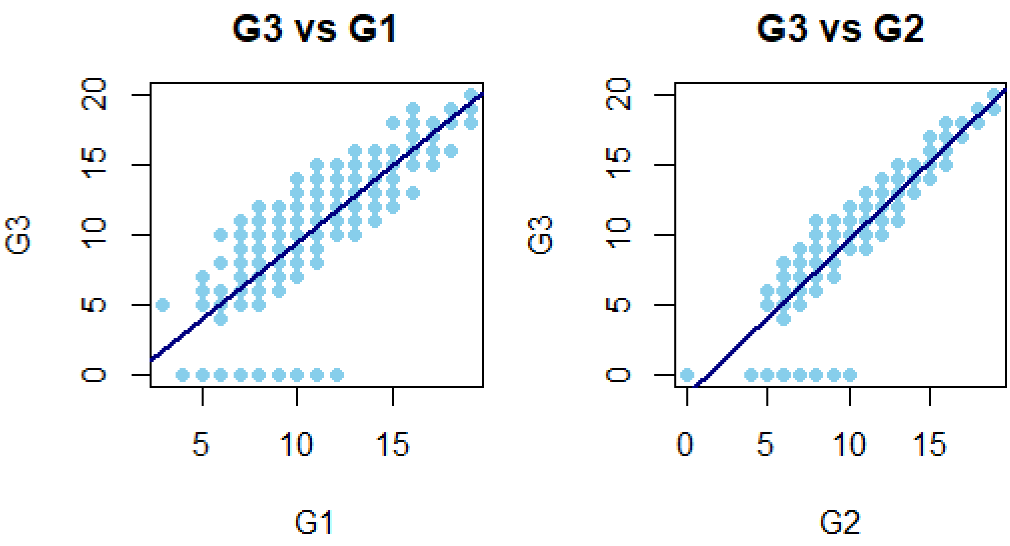
On the other hand, when grades and effort are taken into account, past failures, absences, and most family background factors are not significant, suggesting that their effects are mostly mediated by prior academic success. All things considered, the significant decrease in deviation in comparison to the null model suggests a strong fit and validates the applicability of the logit specification.

We conclude that the probability of passing the final exam is mainly determined by previous grades and individual effort, while the other factors have a limited role once the core variables are taken into account.

```
Call:
glm(formula = PassFail ~ G1 + G2 + failures + absences + studytime +
    sex + age + Medu + Fedu + internet + famsup, family = binomial(link = "logit"),
    data = student_mat)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  9.31392    3.98466   2.337  0.01942 *
G1          -0.44474    0.17551  -2.534  0.01128 *
G2          -1.96386    0.31311  -6.272 3.56e-10 ***
failures    -0.09712    0.30880  -0.315  0.75312
absences     0.03708    0.02966   1.250  0.21125
studytime    0.87402    0.34977   2.499  0.01246 *
sexM         0.59940    0.53498   1.120  0.26254
age          0.51935    0.19700   2.636  0.00838 **
Medu        -0.06006    0.30803  -0.195  0.84541
Fedu         0.64116    0.32899   1.949  0.05131 .
internetyes  0.26072    0.61350   0.425  0.67086
famsupyes    0.21346    0.50220   0.425  0.67080
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 500.50  on 394  degrees of freedom
Residual deviance: 125.28  on 383  degrees of freedom
AIC: 149.28

Number of Fisher Scoring iterations: 8
```

6. Visualizing the results using plots :

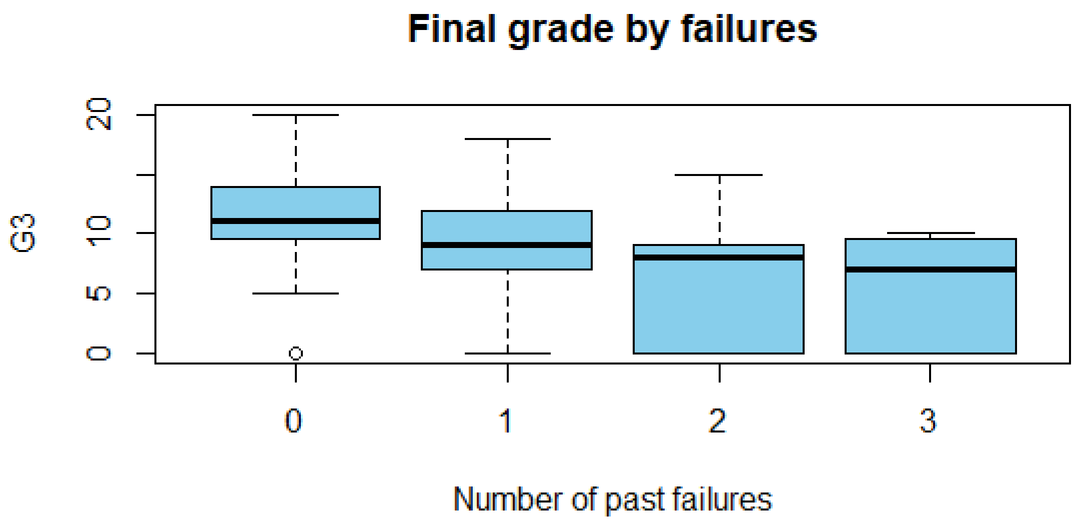
- **G3 vs G1 and G3 vs G2**



The final grade (G3) and previous grades show a strong positive linear relationship, as indicated by both charts. With points more closely matched along the regression line, the connection is clearly closer for G2 than for G1.

Academic achievement is consistent over time; students who do well at the beginning of the year typically do well at the final. The notion that more recent performance provides more pertinent information for forecasting ultimate results is supported by the stronger connection for G2. The latter has the greatest explanatory power in both the SLR and MLR models, as shown in the graph.

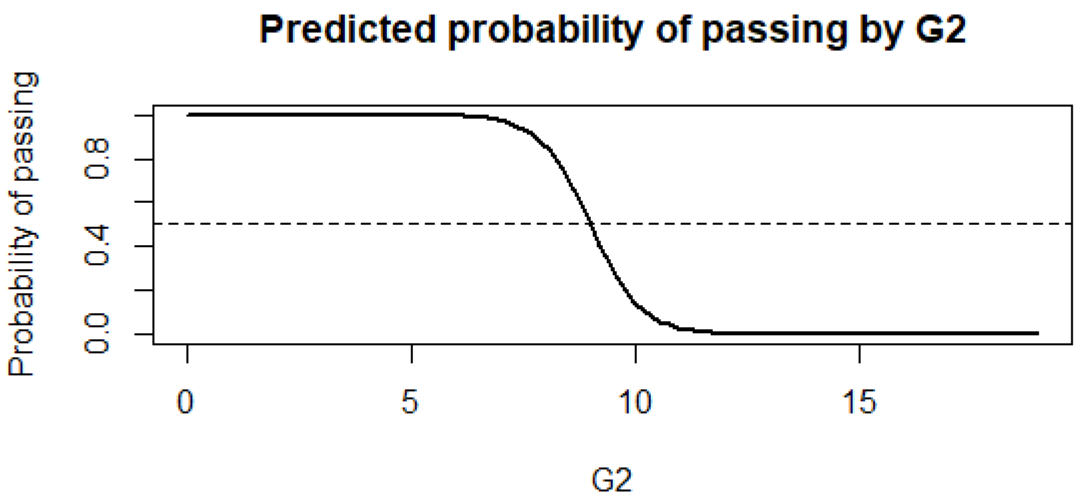
- **G3 by number of failures**



As the number of prior failures rises, the distribution of final grades moves downward. Students who have never failed before have higher median grades and a narrower distribution, whereas those who have failed more than once have lower median grades and greater dispersion.

Past failures indicate ongoing academic challenges that will have a detrimental impact on future performance. The strong negative coefficient for failures in the basic regressions is explained as follows: While some struggling students recover, many remain at risk of poor results, as indicated by the growing number of failures.

- **Probability of passing as a function of G2**



The expected likelihood of passing rapidly shifts around the passing threshold and rises substantially with G2. The expected value of passing is around zero for low values of G2 and near 1 for higher values.

This S-shaped curve, typical of a logit model, shows how small adjustments in recent academic performance near the threshold can greatly affect the probability of success. It validates that G2 is the primary factor determining passing, supporting the use of a binary-choice model rather than linear regression to examine pass-fail outcomes.

Economic interpretation and intuition of the results :

- **Academic Performance Persistence**

The findings show substantial persistence in academic performance over time. Numerous unobserved aspects, including cognitive capacity, learning proficiency, motivation, and discipline, are summarized by early and recent grades. As a result, pupils who perform well early on typically keep their advantage, while those who fail at first have a harder time catching up. This persistence explains why academic variables predominate across all regression models and why their inclusion substantially reduces the explanatory power of other factors.

- **The Role of Past Failures**

Past failures and students' performance are negatively correlated, which may indicate ongoing learning gaps or a lack of interest in school. However, when we consider current grades, the direct effect of past failures becomes much weaker. This indicates that progress during the year might help fix old setbacks and failures, primarily affecting outcomes through their impact on earlier grades.

- **Effort, Study Time, and Measurement Limits**

The variable study time does not have a stable impact on students' performance, leading to measurement limitations. The quantity of effort is captured by the variable, but its efficacy and quality are not. Furthermore, reverse causation may occur when weaker students report studying more in response to challenges. Therefore, even while effort is still conceptually crucial for academic performance, study time alone is an inadequate proxy for productive effort.

- **Why Logit Complements Linear Models**

The results we found using the logit model indicate that academic success is not a purely continuous process but a threshold based one. In fact, gains in recent performance (G2) weigh more than the ones in G1. Therefore, even small gains in recent performance can significantly raise students' chances of passing. This nonlinear relationship emphasizes the value of focused effort and support for students at risk of failing and explains why probability-based models complement linear regressions.

Conclusion

In our research, we used RStudio to analyze a real-world educational dataset on student performance in mathematics. We computed data cleansing, performed descriptive statistics, conducted statistical modeling, and generated visualizations. The main findings of our project indicate that second-period grades are the best predictors of both passing rates and final performance. In linear models, factors such as study time and absence were not significant, but effort was crucial for passing probability, highlighting the potential benefits of small changes for borderline students. Failing in past classes negatively affects outcomes, but current grades mitigate the impact, demonstrating that progression can offset earlier challenges. Family background, especially parental education, influences outcomes indirectly. Our study demonstrates the capabilities of tools such as RStudio for data analysis and underscores the importance of selecting appropriate statistical techniques aligned with research objectives. Overall, we conclude that to obtain meaningful results from data analysis, it is crucial to combine programming skills with thorough interpretation.

Initial data manipulation

```
# Useful packages
library(dplyr)

library(readxl)

# 1. Load the data
student_mat <- read_excel("C:/Users/user/OneDrive/Bureau/student_mat.xlsx")
(student_mat)

## # A tibble: 395 × 33
##    school sex     age address famsize Pstatus  Medu  Fedu Mjob     Fjob
reason
##    <chr>  <chr> <dbl> <chr>   <chr>   <chr>   <dbl> <dbl> <chr>    <chr>
<chr>
##  1 GP     F        18 U       GT3     A           4     4 at_home  teach…
course
##  2 GP     F        17 U       GT3     T           1     1 at_home  other
course
##  3 GP     F        15 U       LE3     T           1     1 at_home  other
other
##  4 GP     F        15 U       GT3     T           4     2 health   servi…
home
##  5 GP     F        16 U       GT3     T           3     3 other    other
home
##  6 GP     M        16 U       LE3     T           4     3 services other
reput…
##  7 GP     M        16 U       LE3     T           2     2 other    other
home
##  8 GP     F        17 U       GT3     A           4     4 other    teach…
home
##  9 GP     M        15 U       LE3     A           3     2 services other
home
## 10 GP     M        15 U       GT3     T           3     4 other    other
home
## # ℹ 385 more rows
## # ℹ 22 more variables: guardian <chr>, traveltime <dbl>, studytime <dbl>,
## #   failures <dbl>, schoolsup <chr>, famsup <chr>, paid <chr>,
## #   activities <chr>, nursery <chr>, higher <chr>, internet <chr>,
## #   romantic <chr>, famrel <dbl>, freetime <dbl>, goout <dbl>, Dalc <dbl>,
## #   Walc <dbl>, health <dbl>, absences <dbl>, G1 <dbl>, G2 <dbl>, G3 <dbl>

#2. Check the structure
str(student_mat)

## tibble [395 × 33] (S3: tbl_df/tbl/data.frame)
##  $ school    : chr [1:395] "GP" "GP" "GP" "GP" ...
##  $ sex       : chr [1:395] "F" "F" "F" "F" ...
##  $ age       : num [1:395] 18 17 15 15 16 16 16 17 15 15 ...
```

```
## $ address   : chr [1:395] "U" "U" "U" "U" ...
## $ famsize   : chr [1:395] "GT3" "GT3" "LE3" "GT3" ...
## $ Pstatus   : chr [1:395] "A" "T" "T" "T" ...
## $ Medu      : num [1:395] 4 1 1 4 3 4 2 4 3 3 ...
## $ Fedu      : num [1:395] 4 1 1 2 3 3 2 4 2 4 ...
## $ Mjob      : chr [1:395] "at_home" "at_home" "at_home" "health" ...
## $ Fjob      : chr [1:395] "teacher" "other" "other" "services" ...
## $ reason    : chr [1:395] "course" "course" "other" "home" ...
## $ guardian  : chr [1:395] "mother" "father" "mother" "mother" ...
## $ traveltime: num [1:395] 2 1 1 1 1 1 1 2 1 1 ...
## $ studytime : num [1:395] 2 2 2 3 2 2 2 2 2 2 ...
## $ failures  : num [1:395] 0 0 3 0 0 0 0 0 0 0 ...
## $ schoolsup : chr [1:395] "yes" "no" "yes" "no" ...
## $ famsup    : chr [1:395] "no" "yes" "no" "yes" ...
## $ paid      : chr [1:395] "no" "no" "yes" "yes" ...
## $ activities: chr [1:395] "no" "no" "no" "yes" ...
## $ nursery   : chr [1:395] "yes" "no" "yes" "yes" ...
## $ higher    : chr [1:395] "yes" "yes" "yes" "yes" ...
## $ internet  : chr [1:395] "no" "yes" "yes" "yes" ...
## $ romantic  : chr [1:395] "no" "no" "no" "yes" ...
## $ famrel    : num [1:395] 4 5 4 3 4 5 4 4 4 5 ...
## $ freetime  : num [1:395] 3 3 3 2 3 4 4 1 2 5 ...
## $ goout     : num [1:395] 4 3 2 2 2 2 4 4 4 4 ...
## $ Dalc      : num [1:395] 1 1 2 1 1 1 1 1 1 1 ...
## $ Walc      : num [1:395] 1 1 3 1 2 2 1 1 1 1 ...
## $ health    : num [1:395] 3 3 3 5 5 5 3 1 1 5 ...
## $ absences  : num [1:395] 6 4 10 2 4 10 0 6 0 0 ...
## $ G1        : num [1:395] 5 5 7 15 6 15 12 6 16 14 ...
## $ G2        : num [1:395] 6 5 8 14 10 15 12 5 18 15 ...
## $ G3        : num [1:395] 6 6 10 15 10 15 11 6 19 15 ...

# 3. Quick overview
summary(student_mat)

##     school              sex                 age          address         
##  Length:395         Length:395         Min.   :15.0   Length:395        
##  Class :character   Class :character   1st Qu.:16.0   Class :character  
##  Mode  :character   Mode  :character   Median :17.0   Mode  :character  
##                                        Mean   :16.7                     
##                                        3rd Qu.:18.0                     
##                                        Max.   :22.0                     
##    famsize            Pstatus               Medu            Fedu      
##  Length:395         Length:395         Min.   :0.000   Min.   :0.000  
##  Class :character   Class :character   1st Qu.:2.000   1st Qu.:2.000  
##  Mode  :character   Mode  :character   Median :3.000   Median :2.000  
##                                        Mean   :2.749   Mean   :2.522  
##                                        3rd Qu.:4.000   3rd Qu.:3.000  
##                                        Max.   :4.000   Max.   :4.000  
##      Mjob               Fjob              reason            guardian        
##  Length:395         Length:395         Length:395         Length:395        
```

```
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##    traveltime      studytime       failures       schoolsup
##  Min.   :1.000   Min.   :1.000   Min.   :0.0000   Length:395
##  1st Qu.:1.000   1st Qu.:1.000   1st Qu.:0.0000   Class :character
##  Median :1.000   Median :2.000   Median :0.0000   Mode  :character
##  Mean   :1.448   Mean   :2.035   Mean   :0.3342
##  3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:0.0000
##  Max.   :4.000   Max.   :4.000   Max.   :3.0000
##     famsup              paid            activities          nursery
##  Length:395         Length:395         Length:395         Length:395
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##     higher            internet           romantic             famrel
##  Length:395         Length:395         Length:395         Min.   :1.000
##  Class :character   Class :character   Class :character   1st Qu.:4.000
##  Mode  :character   Mode  :character   Mode  :character   Median :4.000
##                                                            Mean   :3.944
##                                                            3rd Qu.:5.000
##                                                            Max.   :5.000
##     freetime         goout            Dalc            Walc
##  Min.   :1.000   Min.   :2.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:3.000   1st Qu.:4.000   1st Qu.:1.000   1st Qu.:1.000
##  Median :3.000   Median :4.000   Median :1.000   Median :2.000
##  Mean   :3.235   Mean   :3.977   Mean   :1.481   Mean   :2.291
##  3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:2.000   3rd Qu.:3.000
##  Max.   :5.000   Max.   :4.000   Max.   :5.000   Max.   :5.000
##      health         absences            G1              G2
##  Min.   :1.000   Min.   : 0.000   Min.   : 3.00   Min.   : 0.00
##  1st Qu.:3.000   1st Qu.: 0.000   1st Qu.: 8.00   1st Qu.: 9.00
##  Median :4.000   Median : 4.000   Median :11.00   Median :11.00
##  Mean   :3.554   Mean   : 5.709   Mean   :10.91   Mean   :10.71
##  3rd Qu.:5.000   3rd Qu.: 8.000   3rd Qu.:13.00   3rd Qu.:13.00
##  Max.   :5.000   Max.   :75.000   Max.   :19.00   Max.   :19.00
##        G3
##  Min.   : 0.00
##  1st Qu.: 8.00
##  Median :11.00
##  Mean   :10.42
##  3rd Qu.:14.00
##  Max.   :20.00

summary(student_mat$G1)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    3.00    8.00   11.00   10.91   13.00   19.00 
```

```r
summary(student_mat$G2)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    9.00   11.00   10.71   13.00   19.00 
```

```r
summary(student_mat$G3)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    8.00   11.00   10.42   14.00   20.00 
```

```r
# verify missing values and doubles :

# Number of missing values per variable
colSums(is.na(student_mat))
```

```
##     school        sex        age    address    famsize    Pstatus 
Medu 
##          0          0          0          0          0          0 
0 
##       Fedu       Mjob       Fjob     reason   guardian traveltime 
studytime 
##          0          0          0          0          0          0 
0 
##   failures  schoolsup     famsup       paid activities    nursery 
higher 
##          0          0          0          0          0          0 
0 
##   internet   romantic     famrel   freetime      goout       Dalc 
Walc 
##          0          0          0          0          0          0 
0 
##     health   absences         G1         G2         G3 
##          0          0          0          0          0 
```

```r
student_mat <- student_mat %>% filter(complete.cases(.))

# Checking for duplicates and deleting them
nrow(student_mat) - nrow(distinct(student_mat))
```

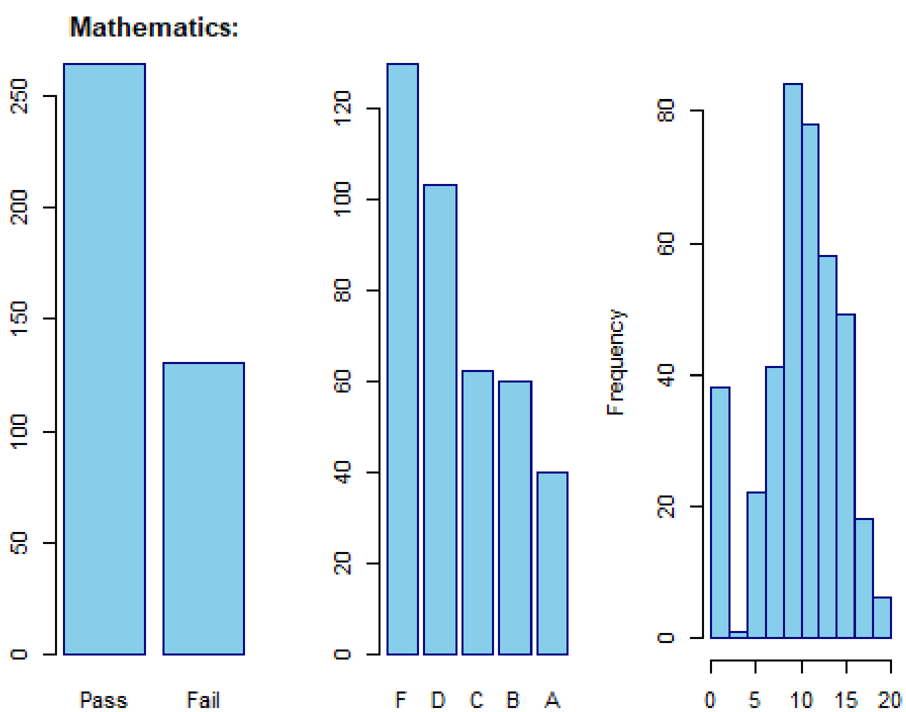
```
## [1] 0
```

```r
student_mat <- distinct(student_mat)

# Extreme values:

student_mat <- student_mat %>% filter(absences > 50)
summary(student_mat$absences)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   54.00   55.00   56.00   61.67   65.50   75.00

# Convert the correct variables into factors

# Categorical variables
cat_variables <- c("school", "sex", "address", "famsize", "Pstatus",
"schoolsup", "famsup", "paid", "activities", "nursery", "higher", "internet",
"romantic")


# Check for the new data
str(student_mat[, cat_variables])

## tibble [3 × 13] (S3: tbl_df/tbl/data.frame)
##  $ school    : chr [1:3] "GP" "GP" "GP"
##  $ sex       : chr [1:3] "F" "F" "F"
##  $ address   : chr [1:3] "U" "U" "R"
##  $ famsize   : chr [1:3] "GT3" "LE3" "GT3"
##  $ Pstatus   : chr [1:3] "T" "T" "A"
##  $ schoolsup : chr [1:3] "yes" "no" "no"
##  $ famsup    : chr [1:3] "yes" "yes" "no"
##  $ paid      : chr [1:3] "yes" "no" "no"
##  $ activities: chr [1:3] "yes" "yes" "no"
##  $ nursery   : chr [1:3] "yes" "yes" "no"
##  $ higher    : chr [1:3] "yes" "yes" "no"
##  $ internet  : chr [1:3] "yes" "yes" "yes"
##  $ romantic  : chr [1:3] "no" "yes" "yes"

#Creating and displaying dummies

student_mat <- read_excel("C:/Users/user/OneDrive/Bureau/student_mat.xlsx")

dummies_Mjob <- model.matrix(~ Mjob - 1, data = student_mat)
head(dummies_Mjob)

##   Mjobat_home Mjobhealth Mjobother Mjobservices Mjobteacher
## 1           1          0         0            0           0
## 2           1          0         0            0           0
## 3           1          0         0            0           0
## 4           0          1         0            0           0
## 5           0          0         1            0           0
## 6           0          0         0            1           0

dummies_Fjob <- model.matrix(~ Fjob - 1, data = student_mat)
head(dummies_Fjob)

##   Fjobat_home Fjobhealth Fjobother Fjobservices Fjobteacher
## 1           0          0         0            0           1
## 2           0          0         1            0           0
## 3           0          0         1            0           0
```

```
## 4           0          0         0            1            0
## 5           0          0         1            0            0
## 6           0          0         1            0            0

dummies_reason <- model.matrix(~ reason - 1, data = student_mat)
head(dummies_reason)

##   reasoncourse reasonhome reasonother reasonreputation
## 1            1          0           0                0
## 2            1          0           0                0
## 3            0          0           1                0
## 4            0          1           0                0
## 5            0          1           0                0
## 6            0          0           0                1

dummies_guardian <- model.matrix(~ guardian - 1, data = student_mat)
head(dummies_guardian)

##   guardianfather guardianmother guardianother
## 1              0              1             0
## 2              1              0             0
## 3              0              1             0
## 4              0              1             0
## 5              1              0             0
## 6              0              1             0

# Manipulating binary variables (pass / fail)
# Create the Pass/Fail binary variable

student_mat <- read_excel("C:/Users/user/OneDrive/Bureau/student_mat.xlsx")

student_mat$PassFail <- ifelse(student_mat$G3 >= 10, "Pass", "Fail")
student_mat$PassFail <- factor(student_mat$PassFail, levels = c("Pass",
"Fail"))
table(student_mat$PassFail)

##
## Pass Fail
##  265  130

prop.table(table(student_mat$PassFail))

##
##      Pass      Fail
## 0.6708861 0.3291139

# 5 categories classification

student_mat$G3_cat5 <- cut(student_mat$G3, breaks = c(-Inf, 9, 11, 13, 15,
20),labels = c("F", "D", "C", "B", "A"), right  = TRUE)
student_mat$G3_cat5 <- factor(student_mat$G3_cat5, levels = c("F", "D", "C",
"B", "A"))
```

```
table(student_mat$G3_cat5)

##
##   F   D   C   B   A
## 130 103  62  60  40

#Plots (3 panels)
op <- par(mfrow = c(1, 3), mar = c(4, 4, 3, 1))

# Panel 1: Pass/Fail barplot
barplot(table(student_mat$PassFail), main = "Mathematics:", col = "skyblue",
border = "navy", ylab = "")

# Panel 2: 5 categories barplot
barplot(table(student_mat$G3_cat5),col = "skyblue",border = "navy",ylab = "")

# Panel 3: Histogram of G3
hist(student_mat$G3,breaks = seq(0, 20, by = 2),col = "skyblue",border =
"navy", xlab = "", main = "")
```




```
par(op)  # restore old plotting settings
```

Regressions and visualizations

```
library(readxl)

# Load the data
student_mat <- read_excel("C:/Users/user/OneDrive/Bureau/student_mat.xlsx")
(student_mat)

## # A tibble: 395 × 33
##    school sex     age address famsize Pstatus  Medu  Fedu Mjob     Fjob
reason
##    <chr>  <chr> <dbl> <chr>   <chr>   <chr>   <dbl> <dbl> <chr>    <chr>
<chr>
##  1 GP     F        18 U       GT3     A           4     4 at_home  teach…
course
##  2 GP     F        17 U       GT3     T           1     1 at_home  other
course
##  3 GP     F        15 U       LE3     T           1     1 at_home  other
other
##  4 GP     F        15 U       GT3     T           4     2 health   servi…
home
##  5 GP     F        16 U       GT3     T           3     3 other    other
home
##  6 GP     M        16 U       LE3     T           4     3 services other
reput…
##  7 GP     M        16 U       LE3     T           2     2 other    other
home
##  8 GP     F        17 U       GT3     A           4     4 other    teach…
home
##  9 GP     M        15 U       LE3     A           3     2 services other
home
## 10 GP     M        15 U       GT3     T           3     4 other    other
home
## # ℹ 385 more rows
## # ℹ 22 more variables: guardian <chr>, traveltime <dbl>, studytime <dbl>,
## #   failures <dbl>, schoolsup <chr>, famsup <chr>, paid <chr>,
## #   activities <chr>, nursery <chr>, higher <chr>, internet <chr>,
## #   romantic <chr>, famrel <dbl>, freetime <dbl>, goout <dbl>, Dalc <dbl>,
## #   Walc <dbl>, health <dbl>, absences <dbl>, G1 <dbl>, G2 <dbl>, G3 <dbl>

# Useful packages
library(dplyr)

vars <- c("sex","school", "address", "famsize","Pstatus", "Mjob","Fjob",
"reason", "guardian", "failures", "schoolsup", "famsup",
"paid","activities","nursery", "higher", "internet", "romantic", "famrel",
"freetime", "age","goout","health",
"absences","G1","G2","G3","studytime","traveltime","Walc","Dalc","Medu","Fedu
")
summary(student_mat[ , vars])
```

```
##      sex                school            address            famsize
##  Length:395         Length:395         Length:395         Length:395
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##    Pstatus             Mjob               Fjob              reason
##  Length:395         Length:395         Length:395         Length:395
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##    guardian           failures       schoolsup            famsup
##  Length:395         Min.   :0.0000   Length:395         Length:395
##  Class :character   1st Qu.:0.0000   Class :character   Class :character
##  Mode  :character   Median :0.0000   Mode  :character   Mode  :character
##                     Mean   :0.3342
##                     3rd Qu.:0.0000
##                     Max.   :3.0000
##      paid            activities          nursery             higher
##  Length:395         Length:395         Length:395         Length:395
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##    internet           romantic             famrel         freetime
##  Length:395         Length:395         Min.   :1.000   Min.   :1.000
##  Class :character   Class :character   1st Qu.:4.000   1st Qu.:3.000
##  Mode  :character   Mode  :character   Median :4.000   Median :3.000
##                                        Mean   :3.944   Mean   :3.235
##                                        3rd Qu.:5.000   3rd Qu.:4.000
##                                        Max.   :5.000   Max.   :5.000
##       age            goout           health          absences
##  Min.   :15.0   Min.   :2.000   Min.   :1.000   Min.   : 0.000
##  1st Qu.:16.0   1st Qu.:4.000   1st Qu.:3.000   1st Qu.: 0.000
##  Median :17.0   Median :4.000   Median :4.000   Median : 4.000
##  Mean   :16.7   Mean   :3.977   Mean   :3.554   Mean   : 5.709
##  3rd Qu.:18.0   3rd Qu.:4.000   3rd Qu.:5.000   3rd Qu.: 8.000
##  Max.   :22.0   Max.   :4.000   Max.   :5.000   Max.   :75.000
##        G1               G2               G3            studytime
##  Min.   : 3.00   Min.   : 0.00   Min.   : 0.00   Min.   :1.000
##  1st Qu.: 8.00   1st Qu.: 9.00   1st Qu.: 8.00   1st Qu.:1.000
##  Median :11.00   Median :11.00   Median :11.00   Median :2.000
##  Mean   :10.91   Mean   :10.71   Mean   :10.42   Mean   :2.035
##  3rd Qu.:13.00   3rd Qu.:13.00   3rd Qu.:14.00   3rd Qu.:2.000
##  Max.   :19.00   Max.   :19.00   Max.   :20.00   Max.   :4.000
##    traveltime          Walc             Dalc             Medu
```

```
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :0.000  
##  1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:2.000  
##  Median :1.000   Median :2.000   Median :1.000   Median :3.000  
##  Mean   :1.448   Mean   :2.291   Mean   :1.481   Mean   :2.749  
##  3rd Qu.:2.000   3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:4.000  
##  Max.   :4.000   Max.   :5.000   Max.   :5.000   Max.   :4.000  
##       Fedu      
##  Min.   :0.000  
##  1st Qu.:2.000  
##  Median :2.000  
##  Mean   :2.522  
##  3rd Qu.:3.000  
##  Max.   :4.000
```

```
# Correlations with G3
numeric_vars <- sapply(student_mat, is.numeric)
cor(student_mat[ , numeric_vars], student_mat$G3, use = "complete.obs")
```

```
##                  [,1]
## age        -0.16157944
## Medu        0.21714750
## Fedu        0.15245694
## traveltime -0.11714205
## studytime   0.09781969
## failures   -0.36041494
## famrel      0.05136343
## freetime    0.01130724
## goout      -0.03290542
## Dalc       -0.05466004
## Walc       -0.05193932
## health     -0.06133460
## absences    0.03424732
## G1          0.80146793
## G2          0.90486799
## G3          1.00000000
```

```
#SLR models

m_mat_G1   <- lm(G3 ~ G1, data = student_mat)
m_mat_G2   <- lm(G3 ~ G2, data = student_mat)
m_mat_medu <- lm(G3 ~ Medu, data = student_mat)
m_mat_fail <- lm(G3 ~ failures, data = student_mat)
m_mat_study <- lm(G3 ~ studytime, data = student_mat)

summary(m_mat_G1)
```

```
## 
## Call:
## lm(formula = G3 ~ G1, data = student_mat)
## 
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max 
## -11.6223  -0.8348   0.3777   1.6965   5.0153 
## 
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)    
## (Intercept) -1.65280    0.47475  -3.481 0.000555 ***
## G1           1.10626    0.04164  26.568  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 2.743 on 393 degrees of freedom
## Multiple R-squared:  0.6424, Adjusted R-squared:  0.6414 
## F-statistic: 705.8 on 1 and 393 DF,  p-value: < 2.2e-16
```

```r
summary(m_mat_G2)
```

```
## 
## Call:
## lm(formula = G3 ~ G2, data = student_mat)
## 
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -9.6284 -0.3326  0.2695  1.0653  3.5759 
## 
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)    
## (Intercept) -1.39276    0.29694   -4.69 3.77e-06 ***
## G2           1.10211    0.02615   42.14  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 1.953 on 393 degrees of freedom
## Multiple R-squared:  0.8188, Adjusted R-squared:  0.8183 
## F-statistic:  1776 on 1 and 393 DF,  p-value: < 2.2e-16
```

```r
summary(m_mat_medu)
```

```
## 
## Call:
## lm(formula = G3 ~ Medu, data = student_mat)
## 
## Residuals:
##      Min       1Q   Median       3Q      Max 
## -11.5517  -1.7342   0.4483   3.2202   9.2658 
## 
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)    
## (Intercept)   7.9167     0.6097   12.98  < 2e-16 ***
## Medu          0.9088     0.2061    4.41 1.34e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 4.478 on 393 degrees of freedom
## Multiple R-squared:  0.04715,    Adjusted R-squared:  0.04473
## F-statistic: 19.45 on 1 and 393 DF,  p-value: 1.336e-05
```

```r
summary(m_mat_fail)
```

```
##
## Call:
## lm(formula = G3 ~ failures, data = student_mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.1572  -2.1572  -0.1572   2.8428   9.0632
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11.1572     0.2361   47.26  < 2e-16 ***
## failures     -2.2204     0.2899   -7.66 1.47e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.279 on 393 degrees of freedom
## Multiple R-squared:  0.1299, Adjusted R-squared:  0.1277
## F-statistic: 58.67 on 1 and 393 DF,  p-value: 1.466e-13
```

```r
summary(m_mat_study)
```

```
##
## Call:
## lm(formula = G3 ~ studytime, data = student_mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.4643  -1.8623   0.5357   3.0697   9.1377
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.3283     0.6033  15.463   <2e-16 ***
## studytime     0.5340     0.2741   1.949   0.0521 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.565 on 393 degrees of freedom
## Multiple R-squared:  0.009569,   Adjusted R-squared:  0.007049
## F-statistic: 3.797 on 1 and 393 DF,  p-value: 0.05206
```

```r
#MLR models

# MLR academic
mlr1_mat <- lm(G3 ~ G1 + G2 + failures + absences + studytime, data =
```

```r
student_mat)
summary(mlr1_mat)

##
## Call:
## lm(formula = G3 ~ G1 + G2 + failures + absences + studytime,
##     data = student_mat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.1894 -0.3662  0.2649  0.9706  3.6031
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.44276    0.43247  -3.336 0.000931 ***
## G1           0.14996    0.05580   2.687 0.007510 **
## G2           0.97726    0.04914  19.888  < 2e-16 ***
## failures    -0.28377    0.14041  -2.021 0.043968 *
## absences     0.03664    0.01205   3.040 0.002530 **
## studytime   -0.17817    0.11717  -1.521 0.129177
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.908 on 389 degrees of freedom
## Multiple R-squared:  0.8287, Adjusted R-squared:  0.8265
## F-statistic: 376.4 on 5 and 389 DF,  p-value: < 2.2e-16

# MLR socio-behavioral
mlr2_mat <- lm(G3 ~ failures + absences + studytime + Walc + Dalc + Medu +
Fedu, data = student_mat)
summary(mlr2_mat)

##
## Call:
## lm(formula = G3 ~ failures + absences + studytime + Walc + Dalc +
##     Medu + Fedu, data = student_mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.0168  -2.0499   0.3525   3.0284   8.1268
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.12142    1.01771   8.963  < 2e-16 ***
## failures    -2.01597    0.30698  -6.567 1.65e-10 ***
## absences     0.02456    0.02732   0.899   0.3694
## studytime    0.18631    0.26823   0.695   0.4877
## Walc         0.06903    0.22333   0.309   0.7574
## Dalc        -0.12196    0.31798  -0.384   0.7015
## Medu         0.62212    0.25554   2.435   0.0154 *
## Fedu        -0.09514    0.25582  -0.372   0.7102
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.257 on 387 degrees of freedom
## Multiple R-squared:  0.1519, Adjusted R-squared:  0.1365
## F-statistic:   9.9 on 7 and 387 DF,  p-value: 2.226e-11
```

```
# MLR socio-economic
mlr3_mat <- lm(G3 ~ Medu + Fedu + famsize + internet + activities , data =
student_mat)
summary(mlr3_mat)
```

```
##
## Call:
## lm(formula = G3 ~ Medu + Fedu + famsize + internet + activities,
##     data = student_mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.2011  -2.2011   0.5187   2.8312   9.3543
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     7.0818     0.7908   8.955  < 2e-16 ***
## Medu            0.7956     0.2668   2.982  0.00305 **
## Fedu            0.1405     0.2655   0.529  0.59707
## famsizeLE3      0.9242     0.4975   1.858  0.06399 .
## internetyes     0.6917     0.6161   1.123  0.26222
## activitiesyes  -0.1005     0.4537  -0.221  0.82486
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.472 on 389 degrees of freedom
## Multiple R-squared:  0.0592, Adjusted R-squared:  0.0471
## F-statistic: 4.895 on 5 and 389 DF,  p-value: 0.0002342
```

```
# Interaction term
mlr5 <- lm(G3 ~ G1 + failures + G1:failures + Medu + studytime, data =
student_mat)
summary (mlr5)
```

```
##
## Call:
## lm(formula = G3 ~ G1 + failures + G1:failures + Medu + studytime,
##     data = student_mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.6825  -0.8219   0.2678   1.6223   5.2354
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.17587    0.69370  -1.695   0.0909 .
## G1           1.07990    0.04761  22.681   <2e-16 ***
## failures     0.08559    0.60347   0.142   0.8873
## Medu         0.16333    0.13020   1.254   0.2104
## studytime   -0.22461    0.16679  -1.347   0.1789
## G1:failures -0.07570    0.07067  -1.071   0.2848
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.715 on 389 degrees of freedom
## Multiple R-squared:  0.6534, Adjusted R-squared:  0.6489
## F-statistic: 146.7 on 5 and 389 DF,  p-value: < 2.2e-16
```

```
# LOGIT model

student_mat$PassFail <- ifelse(student_mat$G3 >= 10, "Pass", "Fail")
student_mat$PassFail <- factor(student_mat$PassFail, levels = c("Pass",
"Fail"))
table(student_mat$PassFail)
```

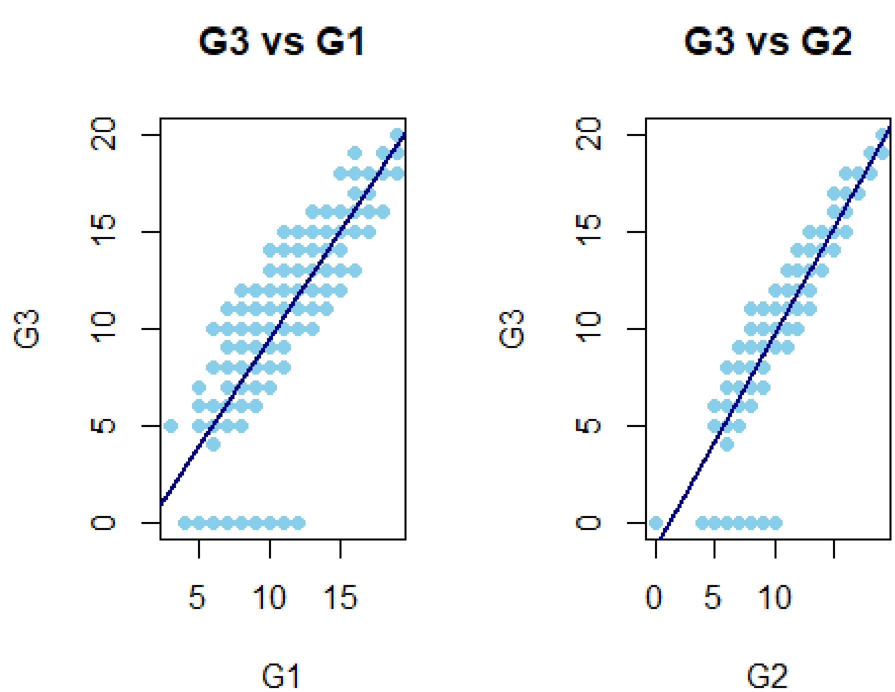
```
##
## Pass Fail
##  265  130
```

```
logit_model <- glm(PassFail ~ G1 + G2 + failures + absences + studytime + sex
+ age + Medu + Fedu + internet + famsup, data = student_mat, family =
binomial(link = "logit"))
summary(logit_model)
```

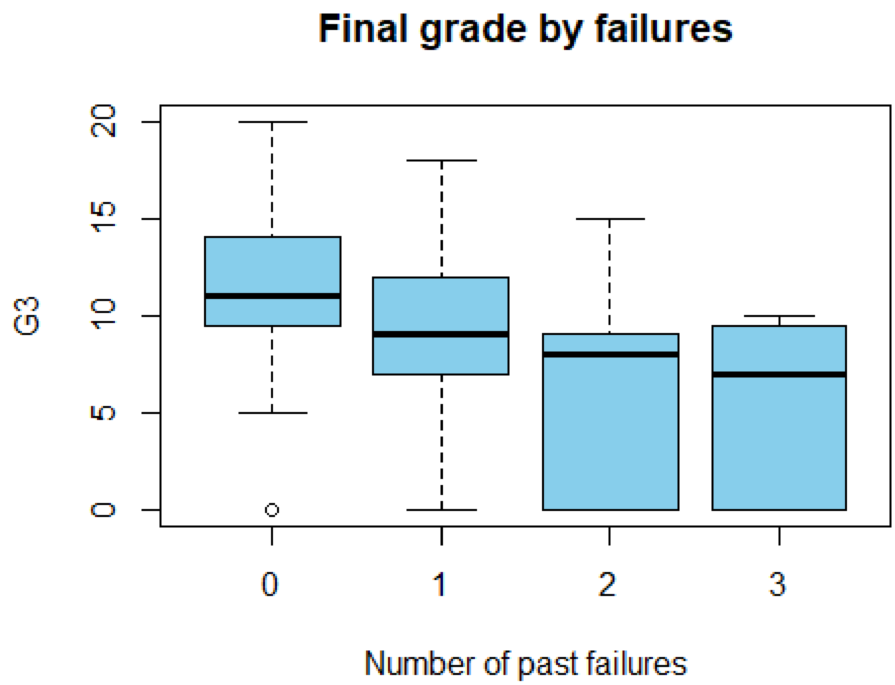
```
##
## Call:
## glm(formula = PassFail ~ G1 + G2 + failures + absences + studytime +
##     sex + age + Medu + Fedu + internet + famsup, family = binomial(link =
"logit"),
##     data = student_mat)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  9.31392    3.98466   2.337  0.01942 *
## G1          -0.44474    0.17551  -2.534  0.01128 *
## G2          -1.96386    0.31311  -6.272 3.56e-10 ***
## failures    -0.09712    0.30880  -0.315  0.75312
## absences     0.03708    0.02966   1.250  0.21125
## studytime    0.87402    0.34977   2.499  0.01246 *
## sexM         0.59940    0.53498   1.120  0.26254
## age          0.51935    0.19700   2.636  0.00838 **
## Medu        -0.06006    0.30803  -0.195  0.84541
## Fedu         0.64116    0.32899   1.949  0.05131 .
## internetyes  0.26072    0.61350   0.425  0.67086
## famsupyes    0.21346    0.50220   0.425  0.67080
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 500.50  on 394  degrees of freedom
## Residual deviance: 125.28  on 383  degrees of freedom
## AIC: 149.28
##
## Number of Fisher Scoring iterations: 8
```

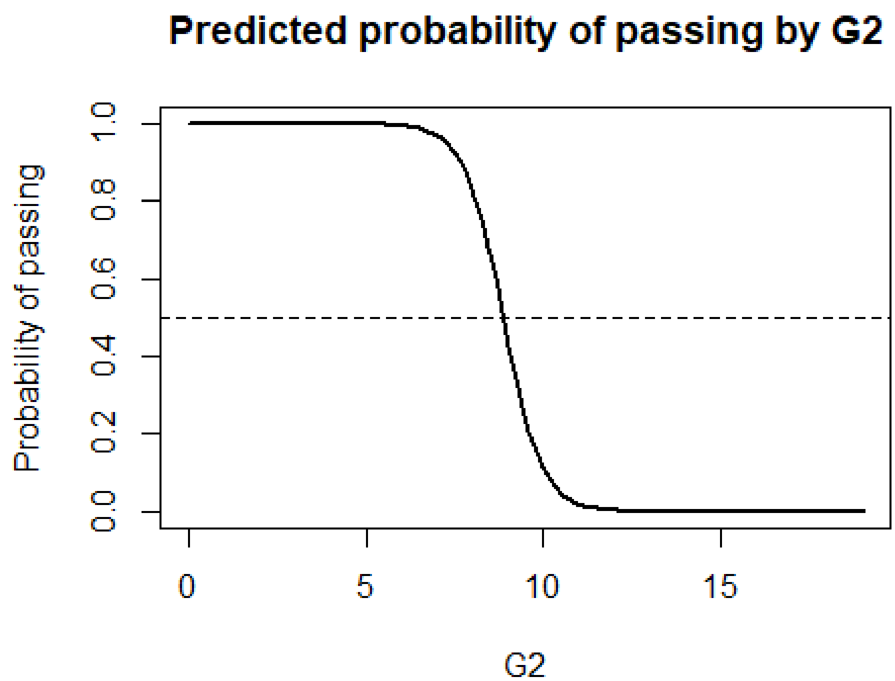
```r
#Visualizing the results using plots :

par(mfrow = c(1,2))

#G3 vs G1

plot(student_mat$G1, student_mat$G3,xlab = "G1", ylab = "G3",main = "G3 vs
G1", pch = 16, col = "skyblue")
abline(lm(G3 ~ G1, data = student_mat), col = "navy", lwd = 2)

#G3 vs G2

plot(student_mat$G2, student_mat$G3,xlab = "G2", ylab = "G3", main = "G3 vs
G2", pch = 16, col = "skyblue")
abline(lm(G3 ~ G2, data = student_mat), col = "navy", lwd = 2)
```



```
par(mfrow = c(1,1))

#G3 by number of failures

boxplot(G3 ~ failures, data = student_mat, col = "skyblue", xlab = "Number of
past failures", ylab = "G3", main = "Final grade by failures")
```




```
#Probability of passing as a function of G2

model_logit <- glm(PassFail ~ G1 + G2 + failures + absences + studytime, data
= student_mat, family = binomial)

G2_seq <- seq(min(student_mat$G2), max(student_mat$G2), length.out = 100)

newdata <- data.frame(G2 = G2_seq, G1 = mean(student_mat$G1), failures = 0,
absences = mean(student_mat$absences), studytime =
mean(student_mat$studytime))

pred_prob <- predict(model_logit, newdata = newdata, type = "response")

plot(G2_seq, pred_prob, type = "l", lwd = 2, xlab = "G2", ylab = "Probability
of passing", main = "Predicted probability of passing by G2")
abline(h = 0.5, lty = 2)
```


Predicted probability of passing by G2
Probability of passing
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
G2